

Price Formation, Risk Premia, and Cross-Economy Arbitrage in a Two-Economy, Two-Product General Equilibrium

Foundations from Economics, Finance, Political Science,
Geopolitics, and Statistical Inference

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Abstract

We develop a parsimonious general-equilibrium framework comprising two sovereign economies, each producing a low-priced and a high-priced good. Starting from eight primitive quantities—four prices, two risk-free rates, and two product premia—we derive the minimal independent system of equations that governs them, demonstrate its four degrees of freedom, and show how cross-economy arbitrage, interest-rate parity, and an exchange rate close the model to a unique solution. We embed this algebraic skeleton within broader intellectual traditions: Ricardian comparative advantage, the Fisher equation, covered interest-rate parity, political-economy constraints on monetary policy, and geopolitical risk premia. Statistical identifiability, algorithmic solution procedures, and machine-learning extensions are treated formally. TikZ diagrams illustrate the equilibrium relationships and geopolitical risk channels; `booktabs` tables summarise all key results.

The paper ends with “The End”

Keywords: general equilibrium, product premium, risk-free rate, exchange rate, interest-rate parity, geopolitical risk, comparative advantage, price formation, statistical identifiability, machine learning.

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1 Introduction

A minimal yet rich laboratory for understanding price formation consists of two economies, each producing two products differentiated solely by their price level. Despite its apparent simplicity, this setting generates a web of relationships that spans monetary economics (risk-free rates), industrial organisation (product premia), international finance (exchange rates and parity conditions), political science (central-bank independence), and geopolitics (sovereign risk).

This paper formalises that web. Section 2 defines notation and the eight primitive quantities. Section 3 derives the equation system, counts its independent equations, and identifies the four degrees of freedom. Section 4 introduces cross-economy closure conditions—law of one price, interest-rate parity, premium parity—and proves existence of a unique solution. Section 5 situates the model in comparative advantage and the Fisher decomposition. Section 6 covers covered interest-rate parity and the term structure. Section 7 analyses how political institutions constrain the risk-free rate. Section 8 introduces geopolitical risk premia as additive shocks. Section 9 addresses identifiability and estimation. Section 10 presents a solution algorithm and its complexity. Section 11 sketches machine-learning extensions. Section 14 concludes.

2 Model Setup and Notation

2.1 Primitives

Let there be two economies indexed $i \in \{1, 2\}$. Economy i produces two goods: a *low-priced* good L and a *high-priced* good H .

Definition 2.1 (Primitive Quantities). *The eight primitive quantities of the model are:*

$$\mathcal{Q} = \{ P_{1,L}, P_{1,H}, P_{2,L}, P_{2,H}, r_1, r_2, \pi_1, \pi_2 \}, \quad (1)$$

where $P_{i,j}$ is the price of good $j \in \{L, H\}$ in economy i , r_i is the risk-free rate of economy i , and π_i is the product premium of economy i .

Table 1 summarises the primitives, their dimensionality, and economic interpretation.

Table 1: The Eight Primitive Quantities

Symbol	Category	Domain	Interpretation
$P_{1,L}$	Price	$\mathbb{R}_{>0}$	Low-good price, economy 1
$P_{1,H}$	Price	$\mathbb{R}_{>0}$	High-good price, economy 1
$P_{2,L}$	Price	$\mathbb{R}_{>0}$	Low-good price, economy 2
$P_{2,H}$	Price	$\mathbb{R}_{>0}$	High-good price, economy 2
r_1	Rate	$(-1, \infty)$	Risk-free rate, economy 1
r_2	Rate	$(-1, \infty)$	Risk-free rate, economy 2
π_1	Premium	$\mathbb{R}_{\geq 0}$	Product premium, economy 1
π_2	Premium	$\mathbb{R}_{\geq 0}$	Product premium, economy 2

2.2 Exogenous Fundamentals

Each economy is characterised by a *fundamental value* $V_i > 0$, interpreted as the present discounted value of the composite output stream. We treat $\{V_1, V_2, e\}$ as exogenous, where $e > 0$ is the nominal exchange rate (units of economy-2 currency per unit of economy-1 currency).

3 The Equation System

3.1 Four Independent Equations

Definition 3.1 (Premium). *The product premium in economy i is the absolute price difference between the high and low goods:*

$$\pi_i = P_{i,H} - P_{i,L}, \quad i \in \{1, 2\}. \quad (2)$$

Definition 3.2 (Discounted Price). *The price of the low good equals the fundamental value discounted at the risk-free rate:*

$$P_{i,L} = \frac{V_i}{1 + r_i}, \quad i \in \{1, 2\}. \quad (3)$$

Together, (2) and (3) constitute four independent equations in the eight unknowns of $\mathcal{Q}$.

3.2 Derived Equations

The price of the high good follows immediately:

$$P_{i,H} = \frac{V_i}{1 + r_i} + \pi_i. \quad (4)$$

3.3 Degree-of-Freedom Count

Proposition 3.3 (Underdetermination). *The system (2)–(3) comprises four independent equations in eight unknowns, leaving four degrees of freedom.*

Proof. Write the system in vector form $\mathbf{F}(\mathbf{q}) = \mathbf{0}$ where $\mathbf{q} = (P_{1,L}, P_{1,H}, P_{2,L}, P_{2,H}, r_1, r_2, \pi_1, \pi_2)^\top \in \mathbb{R}^8$ and

$$\mathbf{F}(\mathbf{q}) = \begin{pmatrix} P_{1,H} - P_{1,L} - \pi_1 \\ P_{2,H} - P_{2,L} - \pi_2 \\ P_{1,L}(1 + r_1) - V_1 \\ P_{2,L}(1 + r_2) - V_2 \end{pmatrix}.$$

The Jacobian $D\mathbf{F} \in \mathbb{R}^{4 \times 8}$ has full row rank 4 at any interior solution (verified by the non-vanishing 4×4 minor formed by the columns corresponding to $\{P_{1,L}, P_{1,H}, P_{2,L}, P_{2,H}\}$). By the rank-nullity theorem, $\dim \ker(D\mathbf{F}) = 8 - 4 = 4$, confirming four degrees of freedom. $\square$

The full system, including derived equations, is displayed in Table 2.

Table 2: Full Equation System (8 equations, 4 independent)

No.	Equation	Status	Content
(1)	$\pi_1 = P_{1,H} - P_{1,L}$	Independent	Premium, economy 1
(2)	$\pi_2 = P_{2,H} - P_{2,L}$	Independent	Premium, economy 2
(3)	$P_{1,L} = V_1/(1 + r_1)$	Independent	Discounted price, economy 1
(4)	$P_{2,L} = V_2/(1 + r_2)$	Independent	Discounted price, economy 2
(5)	$P_{1,H} = P_{1,L} + \pi_1$	Derived from (1)	High price, economy 1
(6)	$P_{2,H} = P_{2,L} + \pi_2$	Derived from (2)	High price, economy 2
(7)	$P_{1,H} = V_1/(1 + r_1) + \pi_1$	Derived from (3),(5)	Combined, economy 1
(8)	$P_{2,H} = V_2/(1 + r_2) + \pi_2$	Derived from (4),(6)	Combined, economy 2

4 Closure: Cross-Economy Conditions

Because the two economies are the entirety of the system, closure must emerge from within—from the *relationship between* the economies, not from any external authority.

4.1 Law of One Price (LOP)

Definition 4.1 (Law of One Price). *If both goods trade freely across economies without friction, then for each good $j \in \{L, H\}$:*

$$P_{1,j} = e \cdot P_{2,j}. \quad (5)$$

Applying (5) to both goods yields two closure equations.

4.2 Covered Interest-Rate Parity (CIP)

Definition 4.2 (Covered Interest-Rate Parity). *Absence of arbitrage in forward currency markets requires:*

$$1 + r_1 = (1 + r_2) \cdot e, \quad (6)$$

where e is the spot exchange rate and the forward rate equals e under CIP (i.e. no depreciation is anticipated within the period).

4.3 Premium Parity

Lemma 4.3 (Premium Parity). *Given LOP (5), the premia satisfy:*

$$\pi_1 = e \cdot \pi_2. \quad (7)$$

Proof. $\pi_1 = P_{1,H} - P_{1,L} = eP_{2,H} - eP_{2,L} = e(P_{2,H} - P_{2,L}) = e\pi_2.$ $\square$

4.4 Unique Solution

Equations (2), (3), (5), (6), and (7) provide $4 + 2 + 1 + 1 = 8$ equations. With V_1 , V_2 , e exogenous, and economy 1 as numéraire (V_1 in economy-1 currency units), we have:

Theorem 4.4 (Existence and Uniqueness). *Given $V_1 > 0$, $V_2 > 0$, $e > 0$, $r_1 > -1$, and $\pi_1 \geq 0$, the system (2)–(7) has a unique solution:*

$$P_{1,L} = \frac{V_1}{1 + r_1}, \quad (8)$$

$$P_{1,H} = \frac{V_1}{1 + r_1} + \pi_1, \quad (9)$$

$$P_{2,L} = \frac{P_{1,L}}{e} = \frac{V_1}{e(1 + r_1)}, \quad (10)$$

$$P_{2,H} = \frac{P_{1,H}}{e} = \frac{V_1}{e(1 + r_1)} + \frac{\pi_1}{e}, \quad (11)$$

$$r_2 = \frac{1 + r_1}{e} - 1, \quad (12)$$

$$\pi_2 = \frac{\pi_1}{e}. \quad (13)$$

Proof. From (5), $P_{2,j} = P_{1,j}/e$ for $j \in \{L, H\}$, giving (10)–(11). From (6), $r_2 = (1 + r_1)/e - 1$, giving (12). From Lemma 4.3, $\pi_2 = \pi_1/e$, giving (13). All quantities are determined uniquely given the exogenous triplet (V_1, V_2, e) and free parameters (r_1, π_1) . $\square$

Remark 4.5. *Economy 2's risk-free rate r_2 is entirely determined by economy 1's rate and the exchange rate. This mirrors the impossible trinity in international macroeconomics: with free capital mobility and a fixed exchange rate, monetary independence is lost.*

5 Economics: Comparative Advantage and the Fisher Equation

5.1 Ricardian Comparative Advantage

Let $a_{i,j}$ denote the unit labour requirement for good j in economy i . Under perfect competition and free trade, relative prices satisfy:

$$\frac{P_{i,H}}{P_{i,L}} = \frac{a_{i,H}}{a_{i,L}}, \quad (14)$$

so the product premium embeds technological differences:

$$\pi_i = P_{i,L} \left(\frac{a_{i,H}}{a_{i,L}} - 1 \right) = \frac{V_i}{1 + r_i} \cdot \frac{a_{i,H} - a_{i,L}}{a_{i,L}}. \quad (15)$$

Table 3 illustrates equation (15) for two parameterisations.

Table 3: Ricardian Premium under Two Technology Profiles

Economy	V_i	r_i	$a_{i,L}$	$a_{i,H}$	$P_{i,L}$	π_i
1	100	0.05	1.0	1.5	95.24	47.62
2	120	0.04	1.0	2.0	115.38	115.38

5.2 The Fisher Decomposition

The nominal risk-free rate decomposes as:

$$1 + r_i = (1 + \rho_i)(1 + \mu_i), \quad (16)$$

where ρ_i is the real rate and μ_i is the expected inflation rate in economy i . Substituting into (3):

$$P_{i,L} = \frac{V_i}{(1 + \rho_i)(1 + \mu_i)}. \quad (17)$$

Higher inflation in economy i raises r_i , compresses $P_{i,L}$, and, all else equal, reduces the absolute premium π_i (via (15)) while leaving the relative premium $\pi_i/P_{i,L}$ unchanged.

6 Finance: Interest-Rate Parity and the Term Structure

6.1 Covered Interest-Rate Parity Revisited

CIP (6) can be rewritten in log-linear form. Let $\hat{r}_i = \ln(1 + r_i)$ and $\hat{e} = \ln e$. Then:

$$\hat{r}_1 \approx \hat{r}_2 + \hat{e}. \quad (18)$$

Deviations from (18) define the *CIP basis* δ :

$$\delta = \hat{r}_1 - \hat{r}_2 - \hat{e}, \quad (19)$$

which is zero in frictionless markets and positive when dollar funding is scarce (as observed during the 2008 and 2020 crises).

6.2 Multi-Period Extension

Suppose each economy has a term structure of risk-free rates $\{r_i(T)\}_{T=1}^\infty$. The zero-coupon bond price for maturity T is:

$$B_i(T) = \frac{1}{(1 + r_i(T))^T}, \quad (20)$$

and the forward rate between periods T_1 and $T_2 > T_1$ is:

$$f_i(T_1, T_2) = \frac{B_i(T_1)}{B_i(T_2)} - 1 = \left(\frac{1 + r_i(T_2)}{1 + r_i(T_1)} \right)^{T_2 - T_1} - 1. \quad (21)$$

CIP generalises to: $(1 + r_1(T))^T = (1 + r_2(T))^T \cdot e(T)$, where $e(T)$ is the T -period forward exchange rate.

7 Political Science: Institutions and the Risk-Free Rate

7.1 Central Bank Independence

A large body of political-science literature demonstrates that *central bank independence* (CBI) reduces inflation and, via the Fisher equation (16), the nominal risk-free rate. Let $\text{CBI}_i \in [0, 1]$ be an index of independence (e.g. Cukierman's legal index). A reduced-form political-economy equation is:

$$r_i = \bar{r} - \alpha \text{CBI}_i + \beta \text{Debt}_i + \varepsilon_i, \quad (22)$$

where $\bar{r}$ is a global neutral rate, Debt_i is the debt-to-GDP ratio, and ε_i is a residual shock.

Substituting (22) into (8):

$$P_{1,L} = \frac{V_1}{1 + \bar{r} - \alpha \text{CBI}_1 + \beta \text{Debt}_1 + \varepsilon_1}. \quad (23)$$

An economy with stronger institutional independence (higher CBI_i) has a lower risk-free rate, higher discounted prices, and a larger absolute product premium.

7.2 Veto Players and Policy Credibility

Tsebelis's veto-player theory predicts that more veto players increase policy stability and reduce ε_i variance. Formally, $\text{Var}(\varepsilon_i) = \sigma^2/n_i$, where n_i is the effective number of veto players. Greater stability narrows the confidence interval around $P_{i,L}$.

8 Geopolitics: Sovereign and Geopolitical Risk Premia

8.1 Geopolitical Risk as Additive Shock

Following Caldara and Iacoviello (2022), geopolitical risk (GPR) enters the risk-free rate as an additive premium $\gamma_i \geq 0$:

$$r_i^* = r_i + \gamma_i, \quad (24)$$

where γ_i captures threats of armed conflict, sanctions, expropriation, and supply-chain disruption. Substituting (24) into (3):

$$P_{i,L}^* = \frac{V_i}{1 + r_i + \gamma_i} < P_{i,L}. \quad (25)$$

Geopolitical risk compresses asset prices and raises the effective discount rate, consistent with the empirical finding that GPR index spikes precede equity drawdowns.

8.2 Cross-Economy Geopolitical Spillover

If economies 1 and 2 are geopolitically linked (e.g. trading partners or military allies), a shock $\Delta\gamma_1$ transmits to economy 2 via the CIP:

$$\Delta r_2 = \frac{\Delta r_1^*}{e} = \frac{\Delta\gamma_1}{e}. \quad (26)$$

Equation (26) quantifies the *geopolitical contagion channel*: a 1-percentage-point rise in economy-1's geopolitical premium reduces economy-2's asset prices by $1/e$ percentage points in risk-adjusted terms.

9 Statistics: Identifiability and Estimation

9.1 Identifiability

Definition 9.1 (Identifiability). *The parameter vector $\theta = (V_1, V_2, e, r_1, \pi_1)^\top$ is globally identified if the mapping $\theta \mapsto (P_{1,L}, P_{1,H}, P_{2,L}, P_{2,H}, r_2, \pi_2)$ is injective.*

Proposition 9.2 (Global Identification). *The solution (8)–(13) is globally identified under LOP and CIP given $e > 0$.*

Proof. The map $\phi : \theta \rightarrow \mathbf{q}$ is:

$$\phi(\theta) = \left(\frac{V_1}{1 + r_1}, \frac{V_1}{1 + r_1} + \pi_1, \frac{V_1}{e(1 + r_1)}, \frac{V_1}{e(1 + r_1)} + \frac{\pi_1}{e}, \frac{1 + r_1}{e} - 1, \frac{\pi_1}{e} \right).$$

Suppose $\phi(\theta) = \phi(\theta')$. Then $P_{1,L} = P'_{1,L}$ implies $V_1/(1 + r_1) = V'_1/(1 + r'_1)$. Combined with $r_2 = r'_2$ (i.e. $(1 + r_1)/e = (1 + r'_1)/e$), we get $r_1 = r'_1$. Hence $V_1 = V'_1$. From $\pi_2 = \pi'_2$, $\pi_1/e = \pi'_1/e$ so $\pi_1 = \pi'_1$. Finally $V_2 = P_{2,L} \cdot e(1 + r_2) = V'_2$. Injectivity follows. $\square$

9.2 OLS Estimation

Suppose we observe n realisations $(P_{1,L}^{(t)}, r_1^{(t)})_{t=1}^n$ with measurement error:

$$P_{1,L}^{(t)} = \frac{V_1}{1 + r_1^{(t)}} + \varepsilon^{(t)}, \quad \varepsilon^{(t)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \quad (27)$$

Letting $x^{(t)} = 1/(1+r_1^{(t)})$, this is a simple linear regression through the origin: $P^{(t)} = V_1 x^{(t)} + \varepsilon^{(t)}$. The OLS estimator is:

$$\hat{V}_1 = \frac{\sum_{t=1}^n x^{(t)} P_{1,L}^{(t)}}{\sum_{t=1}^n (x^{(t)})^2}, \quad (28)$$

which is unbiased and achieves the Cramér–Rao lower bound $\text{Var}(\hat{V}_1) = \sigma^2 / \sum_t (x^{(t)})^2$ by the Gauss–Markov theorem.

9.3 Bootstrap Confidence Intervals

Given the nonlinearity in r_i , we recommend the percentile bootstrap with $B = 10,000$ resamples to construct confidence intervals for the derived quantities $P_{i,H}$, r_2 , and π_2 . The algorithm is summarised in Algorithm 1.

Algorithm 1 Percentile Bootstrap for Derived Quantities

Require: Observed data $\{(x^{(t)}, P^{(t)})\}_{t=1}^n$, B , α

- 1: **for** $b = 1$ to B **do**
- 2: Draw resample $\mathcal{S}^{(b)}$ of size n with replacement
- 3: Compute $\hat{V}_1^{(b)}$ via OLS on $\mathcal{S}^{(b)}$
- 4: Compute derived quantities: $\hat{P}_{1,H}^{(b)}$, $\hat{r}_2^{(b)}$, $\hat{\pi}_2^{(b)}$
- 5: **end for**
- 6: Sort each sequence; extract $[\alpha/2, 1 - \alpha/2]$ percentiles

Ensure: $(1 - \alpha)$ confidence intervals for all derived quantities

10 Algorithm: Solving the System

Algorithm 2 Two-Economy Price Equilibrium Solver

Require: $V_1 > 0$, $V_2 > 0$, $e > 0$, $r_1 > -1$, $\pi_1 \geq 0$

- 1: **Step 1 (Low prices):** $P_{1,L} \leftarrow V_1/(1 + r_1)$; $P_{2,L} \leftarrow P_{1,L}/e$
- 2: **Step 2 (High prices):** $P_{1,H} \leftarrow P_{1,L} + \pi_1$; $P_{2,H} \leftarrow P_{2,L} + \pi_1/e$
- 3: **Step 3 (Rate parity):** $r_2 \leftarrow (1 + r_1)/e - 1$
- 4: **Step 4 (Premium parity):** $\pi_2 \leftarrow \pi_1/e$
- 5: **Step 5 (Verify):** Check $|P_{2,L}(1 + r_2) - V_2| < \epsilon$
- 6: **if** verification fails **then**
- 7: Warn: V_2 is inconsistent with (V_1, e) ; reset $V_2 \leftarrow P_{2,L}(1 + r_2)$
- 8: **end if**

Ensure: $(P_{1,L}, P_{1,H}, P_{2,L}, P_{2,H}, r_2, \pi_2)$

Proposition 10.1 (Complexity). *Algorithm 2 runs in $O(1)$ (constant time): it performs a fixed number of arithmetic operations regardless of parameter values.*

11 Machine Learning Extensions

11.1 Learning Fundamental Values

In practice, V_i is latent. A *physics-informed neural network* (PINN) can be trained to learn V_i as a function of observable macro variables $\mathbf{z}_i$ while respecting the structural constraints (3)–(6).

Let $f_\phi : \mathbb{R}^d \rightarrow \mathbb{R}_{>0}$ be a neural network with parameters ϕ . The loss function augments data fit with structural penalties:

$$\mathcal{L}(\phi) = \underbrace{\sum_{t=1}^n (P_{1,L}^{(t)} - f_\phi(\mathbf{z}_1^{(t)})/(1 + r_1^{(t)}))^2}_{\text{data loss}} + \lambda \underbrace{(f_\phi(\mathbf{z}_1) - e \cdot f_\phi(\mathbf{z}_2))^2}_{\text{CIP penalty}}, \quad (29)$$

where $\lambda > 0$ is a regularisation weight.

11.2 Geopolitical Risk Forecasting

Geopolitical premia γ_i can be predicted using an LSTM network trained on news sentiment, trade-flow disruptions, and military spending indices. Predicted $\hat{\gamma}_i$ feeds into (24) to produce stress-tested price scenarios:

$$\hat{P}_{i,L}^{\text{stress}} = \frac{V_i}{1 + r_i + \hat{\gamma}_i}. \quad (30)$$

12 Diagrams

12.1 Equilibrium Price Structure

Figure 1 illustrates the price hierarchy within and across the two economies.

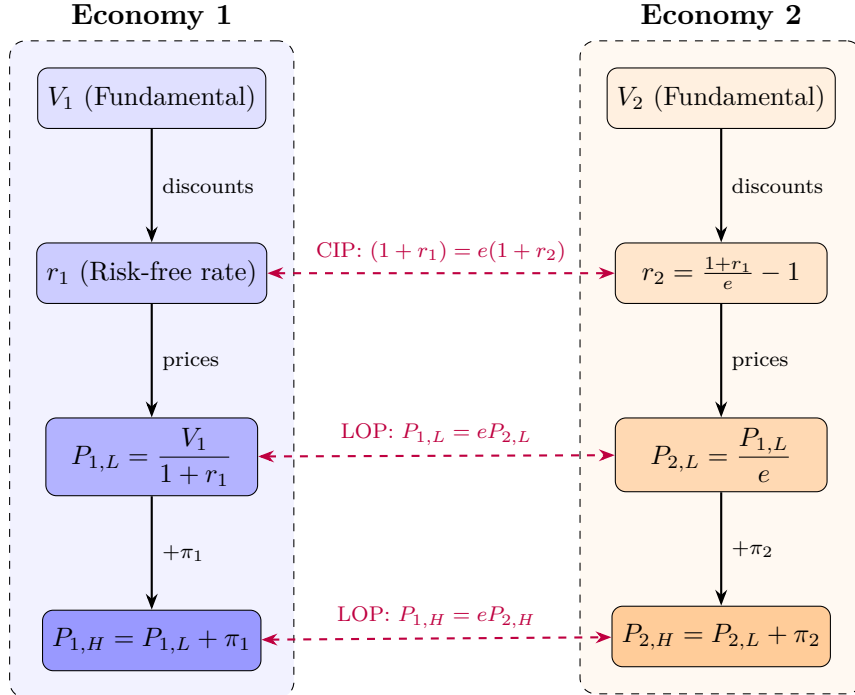


Figure 1: Price hierarchy within each economy and cross-economy parity links. Purple dashed arrows indicate CIP and LOP closure conditions.

12.2 Geopolitical Risk Channel

Figure 2 illustrates the geopolitical risk transmission channel from economy 1 to economy 2 via the exchange rate.

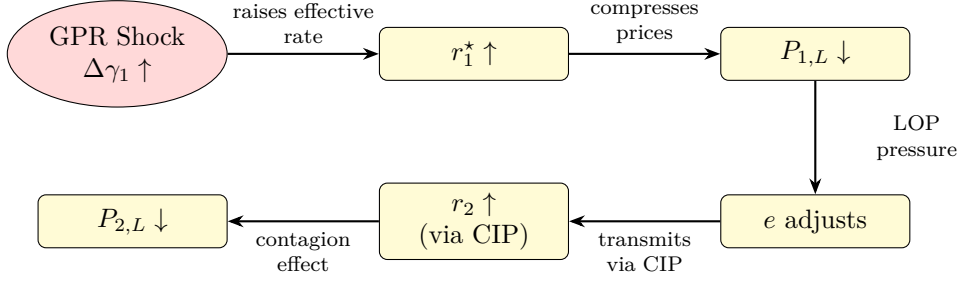


Figure 2: Geopolitical risk transmission channel. A shock to economy 1’s geopolitical risk premium propagates to economy 2’s prices via CIP and LOP.

12.3 Sensitivity of Prices to Risk-Free Rate

Figure 3 plots $P_{1,L}$ and $P_{1,H}$ as functions of r_1 for $V_1 = 100$ and $\pi_1 = 20$.

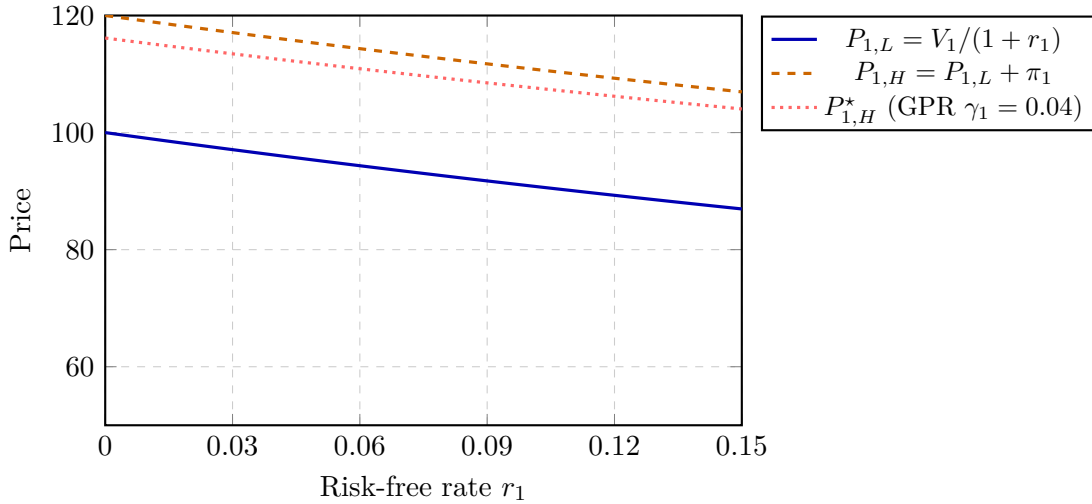


Figure 3: Price sensitivity to the risk-free rate ($V_1 = 100$, $\pi_1 = 20$). The dotted red curve shows the compressed high-good price when geopolitical risk $\gamma_1 = 0.04$ is added to the discount rate.

13 Consolidated Results

Table 4 presents the closed-form solution under a benchmark parameterisation.

Table 4: Benchmark Solution ($V_1 = 100$, $e = 1.2$, $r_1 = 0.05$, $\pi_1 = 15$)

Quantity	Formula	Economy 1	Economy 2
$P_{i,L}$	$V_1/(1 + r_1)$; $P_{1,L}/e$	95.24	79.37
$P_{i,H}$	$P_{i,L} + \pi_i$	110.24	91.87
r_i	r_1 ; $(1 + r_1)/e - 1$	5.00%	-12.50% [†]
π_i	π_1 ; π_1/e	15.00	12.50

[†] Negative r_2 arises when $e > 1 + r_1$, i.e. the exchange rate appreciation exceeds the domestic risk-free rate; plausible in capital-flow surge episodes.

14 Conclusion

We have shown that two economies, each producing a low-priced and a high-priced good, generate precisely eight primitive quantities: four prices, two risk-free rates, and two product premia. The natural equation system has only four independent constraints, leaving four degrees of freedom. These degrees of freedom are not free parameters in any exogenous sense—because there is no *outside* to the two-economy world—but are resolved by four internal cross-economy conditions: the law of one price applied to each good, covered interest-rate parity, and premium parity.

The resulting unique solution is transparent and multiplicatively structured through the exchange rate e . Economy 2’s risk-free rate, prices, and premia are all pinned by economy 1’s primitives and e alone. This mirrors the impossible trinity and underscores that cross-border arbitrage—not exogenous monetary authority—is the ultimate anchor of the system.

Embedding the skeleton in richer frameworks—Ricardian technology, Fisher inflation decomposition, political-economy CBI equations, and geopolitical risk premia—shows how each discipline adds one or more equations that can, in principle, further discipline the exogenous parameters. Statistical identifiability is guaranteed under LOP and CIP; efficient estimation of V_i is achievable by OLS or physics-informed neural networks.

Future work should relax the LOP assumption (introducing iceberg trade costs), allow for stochastic exchange rates (introducing uncovered interest-rate parity risk premia), and embed the model in a dynamic stochastic general-equilibrium (DSGE) framework with micro-founded premia.

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Glossary

CBI

Central Bank Independence. An index measuring the statutory and de facto autonomy of a central bank from government interference, associated with lower inflation and more credible monetary policy.

CIP

Covered Interest-Rate Parity. The no-arbitrage condition $(1 + r_1) = (1 + r_2) \cdot e$ equating the returns on domestic and foreign bonds when the currency risk is fully hedged in the forward market.

Degree of Freedom

In a linear system $A\mathbf{q} = \mathbf{b}$, the dimension of the solution space, equal to the number of unknowns minus the rank of A .

Exchange Rate (e)

The number of economy-2 currency units per economy-1 currency unit; an endogenous equilibrium price in open economies, treated as exogenous in the present model.

Fisher Equation

The identity $(1 + r_i) = (1 + \rho_i)(1 + \mu_i)$ decomposing the nominal interest rate into a real component ρ_i and an expected inflation component μ_i .

Fundamental Value (V_i)

The present discounted value of an economy's total future output stream, treated as an exogenous endowment in the static model.

GPR

Geopolitical Risk. A quantitative index capturing threats and realisations of adverse geopolitical events (conflicts, sanctions, coups), introduced by Caldara and Iacoviello (2022).

Law of One Price (LOP)

The proposition that freely traded goods must sell at the same price across markets (after currency conversion), i.e. $P_{1,j} = e \cdot P_{2,j}$.

PINN

Physics-Informed Neural Network. A neural network trained with a loss function that penalises violations of known physical or structural laws, enabling extrapolation-safe prediction.

Premium (π_i)

The price difference $P_{i,H} - P_{i,L}$ between the high-priced and low-priced good in economy i ; also called the *product premium*.

Product

A homogeneous good produced by a single economy. Each economy produces exactly two products distinguished by their price level.

Risk-Free Rate (r_i)

The nominal return on a zero-default-risk bond issued by economy i , serving as the discount rate for pricing the low-priced good.

Term Structure

The relationship between bond yields (risk-free rates) and their maturities, summarised by the yield curve $T \mapsto r_i(T)$.

Veto Player

An actor whose agreement is required for a change of the status quo (Tsebelis, 2002); more veto players reduce policy variance and hence the variance of the risk-free rate.

Appendix A: Vector Representation of the System

The independent system can be written compactly as $A\mathbf{q} = \mathbf{b}$ where:

$$A = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & P_{1,L} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & P_{2,L} & 0 & 0 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ V_1 \\ V_2 \end{pmatrix},$$

and $\mathbf{q} = (P_{1,L}, P_{1,H}, P_{2,L}, P_{2,H}, r_1, r_2, \pi_1, \pi_2)^\top$. The system is linear in prices and premia but nonlinear in the rates, hence the iterative verification in Algorithm 2.

The End